Assignment 2: Cut (!) and Negation as failure

1. Define the procedure **[10 marks]**

```
split(Numbers, Positives, Negatives)
```

which splits a list of numbers into two lists: positive ones(including zero)a nd negative ones. For example,

```
split([3,-1,0,5,-2],[3,0,5],[ -1,-21)
```

Propose two versions: one with a cut and one without.

2. Consider the following knowledge base: **[10 marks]**

```
directTrain(guwahati, tezpur).

directTrain(nagaon, guwahati).

directTrain(lumding, nagaon).

directTrain(haflong, lumding).

directTrain(silchar, haflong).

directTrain(agartala, silchar).

directTrain(aizawl, agartala).
```

Assume that whenever it is possible to take a direct train from A to B, it is also possible to take a direct train from B to A. Add this information to the database. Then write a predicate `route/3` which gives you a list of towns that are visited by taking the train from one town to another. For instance:

```
?- route(aizawl, guwahati, R).

R = [aizawl, agartala, silchar, haflong, lumding, nagaon,
guwahati].
```

3. The following predicate number_of_parents give information about how many parents somebody has. **[10 marks]**

```
number_of_parents(adam,0):- !.

number_of_parents(eve,0):- !.

number_of_parents(X,2).
```

That is the number of parents is 0 for adam and eve, but 2 for everybody else. First identify the problem in this implementation and modify it to recover from the problem.